

Research Conclusions

Europe's AI boom is not inherently environmentally sustainable. Its sustainability depends primarily on each country's renewable-energy mix and digital infrastructure.

Key Findings

- **Enterprise AI adoption alone is not associated with higher CO₂ emissions.** Countries with higher AI adoption do not necessarily produce more emissions.
- **Data-centre infrastructure shows the strongest relationship with CO₂ emissions.** Countries with more listed data centres generally have higher power-sector emissions, although this is also influenced by country size, electricity demand, industrial activity, and digital-hub status.
- **Renewable energy plays a key role in reducing environmental impact.** Countries with stronger renewable-energy systems generally experience lower emissions and are better positioned for sustainable AI growth.
- **The most sustainable conditions combine AI adoption with high renewable-energy shares.** In contrast, countries with expanding digital infrastructure but weaker renewable-energy systems may face greater environmental pressure.
- **Correlation does not imply causation.** This analysis identifies country-level patterns but does not prove that AI or data centres directly cause higher emissions.

Final Answer

Europe's AI boom can be environmentally sustainable, but only when digital infrastructure expands alongside renewable energy and cleaner electricity systems.